

WEEK 6 PROGRESS SUMMARY

1. Planned to Accomplish

Deepen the reminder-effectiveness analysis and build out the KPI dashboard for stakeholder review, continuing directly from the Week 5 plan.

2. Completed

Instead of the planned reminder-deep-dive, pivoted to understanding *why* no-shows happen rather than just *what* the factors suggested in Week 5. Revised the dashboard from the Week 5 version into a sharper, findings-focused layout (Appointment Outcome, Reminder Status, Distance to Clinic, and Booking Lead Days panels), and scoped the analysis specifically to Attended vs. No-Show comparisons.

3. Improved from Week 5

Week 5's dashboard surfaced factors without interpreting them. The dashboard itself was also simplified, dropping panels that weren't adding interpretive value in favor of ones that support the four advanced findings.

4. What Was Integrated

Feedback from a data science intern was incorporated directly into the dashboard's scope and structure - specifically, the decision to exclude Cancelled appointments so the analysis stays strictly focused on No-Show cases, sharpening the comparison at the center of the dashboard.

5. Track Collaborated With

Data Science.

6. What Was Exchanged

Requested guidance on what additional factors could strengthen the analysis, with the goal of making any future handoff to a data scientist self-explanatory for model-building. In return, provided the data science intern with a specific data-quality flag: `waiting_time_minutes` is populated even for No-Show and Cancelled appointments, which shouldn't logically have a wait time recorded.

7. What Changed as a Result

The dashboard was revised to communicate findings more clearly rather than just display them, and the analysis scope was tightened to exclude Cancelled appointments. A direct product of the collaboration rather than an independent decision.

8. Key Findings or Development Outcomes

- Reminder effectiveness remains modest at 4.03 percentage points
- 72.43% of no-shows come from patients with 2+ prior no-shows, confirming concentration over even distribution.
- Distance shows a measurable separation: Attended patients average 9.67km from the facility versus 10.53km for No-Show patients.
- Booking lead time shows a visible effect. Patients averaging around 34 lead days were more likely to no-show than those around 24 days.

9. Major Challenges

Shifting focus mid-week from the planned reminder/dashboard work to a deeper causal exploration meant the original Week 6 plan wasn't followed as scoped.

10. Important Decisions

Decided to exclude Cancelled appointments from the analysis entirely, treating this as a scope decision rather than a data quality fix, since Cancelled and No-Show represent different operational situations. Also decided to prioritize interpretive depth over

dashboard breadth this week, given the collaboration input suggested the earlier version wasn't yet communicating findings effectively.

11. Remaining Issues

distance_to_clinic_km still has 1.8% missing values, small enough that it likely doesn't bias results in either direction, but not yet resolved. The reminder-effectiveness, distance, and lead-time patterns are still associations pending the Week 7 statistical testing to confirm they're not just sample variation.

12. Contribution to the Overall HealthConnect Project

Delivered the first version of findings that moves beyond description into interpretation — giving the project concrete, stakeholder-ready reasoning (e.g., the mobile clinic and booking cut-off recommendations) rather than just metrics. Also established a working exchange with Data Science, surfacing a data-quality issue early enough that it won't quietly undermine a future model.

13. Proposed Focus for Week 7

Run the three planned statistical tests; chi-square on reminder effectiveness, t-tests on distance and lead-time gaps, and a concentration check on the repeat-offender share to move the Week 6 findings from observed patterns to statistically validated ones.